Q1. As shown in Fig. 1, NMOS transistor M_1 is part of a CS amplifier, and it operates in inversion region.

(1) Derive the expression of transistor intrinsic gain A_v, defined as $g_m r_o$, in terms of bias current I_D and the size W/L of M_1.

(2) Explain why A_v cannot be increased by simply increasing I_D.

(10 Marks)

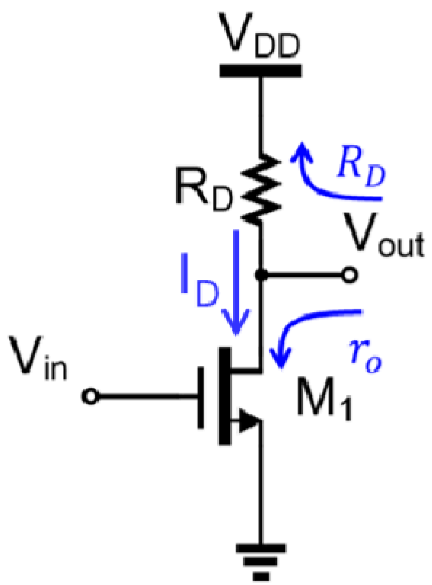
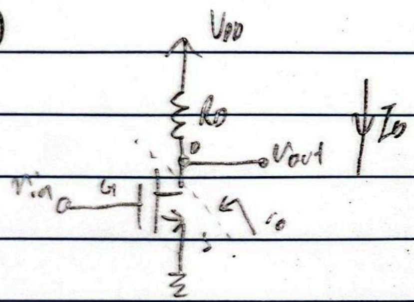


Fig. 1 CS amplifier with resistive load

1)



i) $A_v = g_m r_o$ ← defined by question (even though should have -ve sign)

$= 2\sqrt{\frac{1}{2}\left(\frac{W}{L}\right)k_n' I_D} \times \frac{1}{\lambda I_D}$

$= \sqrt{2\left(\frac{W}{L}\right)k_n' \frac{1}{I_D}} \times \frac{1}{\lambda}$ #

ii) increasing I_D, increases g_m in $\sqrt{I_D}$, but decreases r_o in $\frac{1}{I_D}$

$\therefore$ A_v cannot be incre[a]sed by increasing I_D

Q2. Use small-signal model to derive the equivalent resistance r_{eq} looking to the transistor terminal as shown in Fig.2 (a) and Fig. 2 (b) below. (20 Marks)

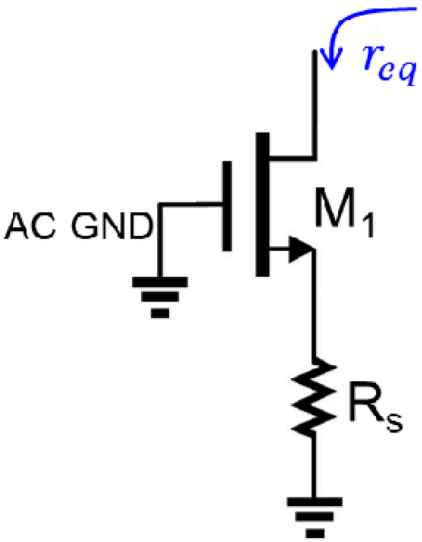


Fig. 2 (a)

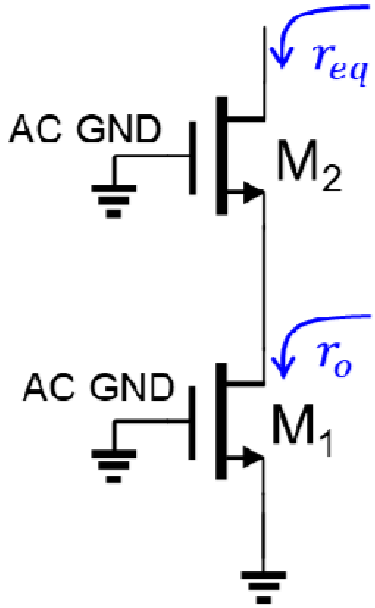
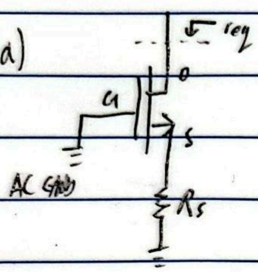


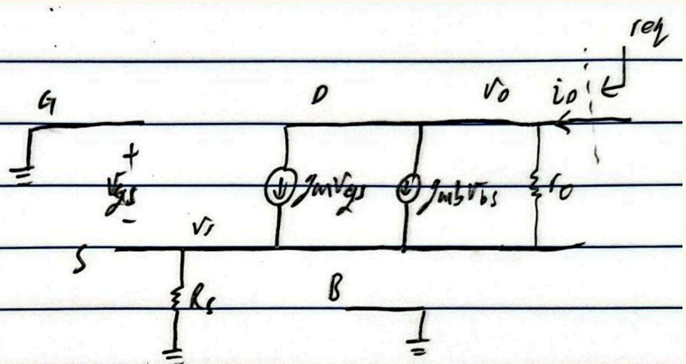
Fig. 2 (b)

2)a)



Assume in saturation

Assume body is grounded



$v_{gs} = v_{bs} = 0 - v_s = -v_s$

KCL at D: $i_o = g_m v_{gs} + g_{mb} v_{bs} + \frac{v_o - v_s}{r_o} = v_s / R_s$ $\qquad r_{eq} = \frac{\partial v_o}{\partial i_o}$

$i_o = -(g_m + g_{mb}) v_s + \frac{v_o - v_s}{r_o} = \frac{v_s}{R_s}$

$\downarrow \partial v_o$

$\frac{\partial i_o}{\partial v_o} = -(g_m + g_{mb}) \frac{\partial v_s}{\partial v_o} + \frac{1}{r_o} - \frac{1}{r_o}\frac{\partial v_s}{\partial v_o} = \frac{\partial v_s}{\partial v_o}\frac{1}{R_s}$

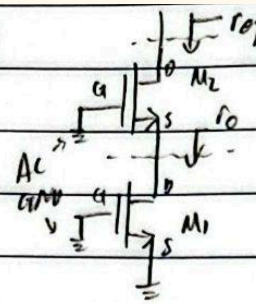
$\frac{\partial v_s}{\partial v_o} = \frac{\frac{1}{r_o}}{\frac{1}{R_s} + (g_m + g_{mb}) + \frac{1}{r_o}}$

$r_{eq} = \frac{\partial v_o}{\partial i_o} = R_s \times \frac{\partial v_o}{\partial v_s}$

$= R_s + r_o + R_s r_o (g_m + g_{mb})$ #

$= \frac{R_s}{R_s + r_o + R_s r_o (g_m + g_{mb})}$

2b)



from part a),

$r_{eq} = r_o + r_{o2} + r_o r_{o2}(g_{m2}+g_{mb2})$ #

Q3. Fig. 3 shows a common source amplifier with source degeneration resistor R_S. Derive the expression for amplifier voltage gain A_v. (20 Marks)

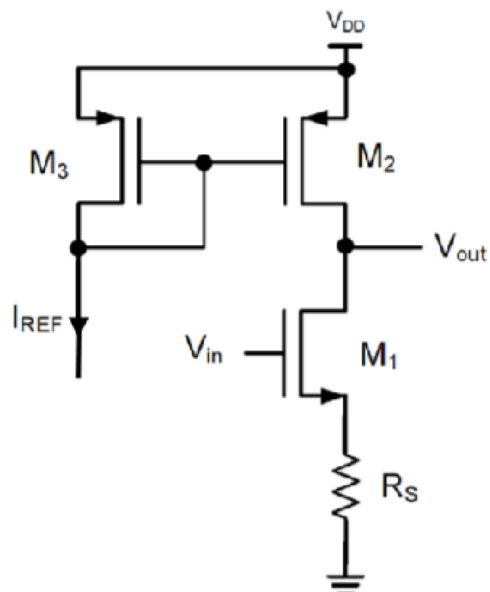
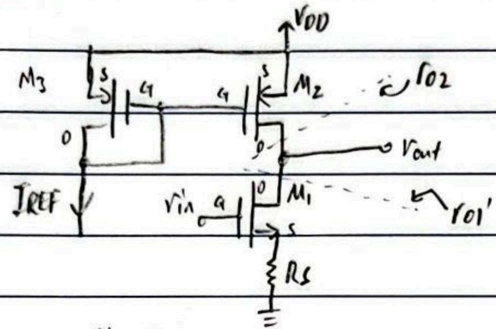
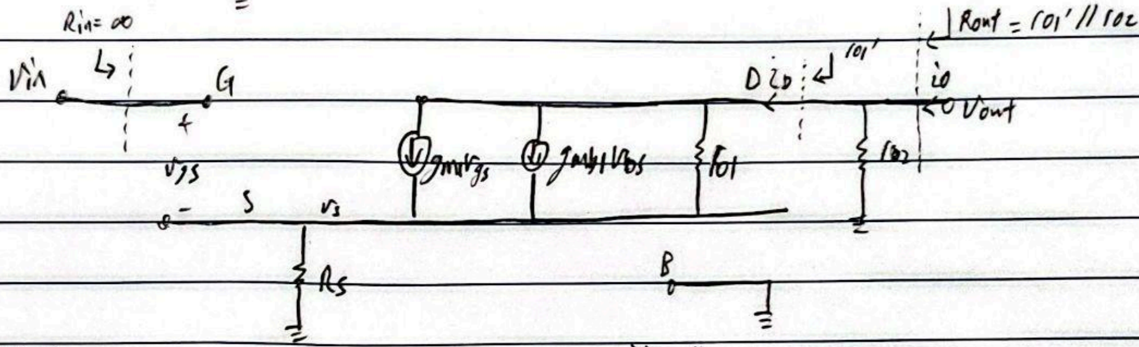


Fig. 3 CS amplifier with CSL and source degeneration resistor

3)



Assume body is tied to GND for M_1 & V_{DD} for $M_2 M_3$



$r_{o1}' = \frac{\partial v_{out}}{\partial i_o} \Big|_{v_{in}=0}$

from 2a) $\Rightarrow$ $r_{o1}' = R_s + r_{o1} + R_s r_{o1}(g_{m1} + g_{mb1})$

$G_M = \frac{\partial i_o}{\partial v_{in}} \Big|_{v_{out}=0}$

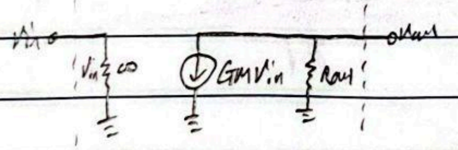
$i_o = g_{m1}v_{gs} + g_{mb1}v_{bs} + \frac{V_{out} - v_s}{r_{o1}} + \frac{V_{out}}{r_{o2}}$

$i_o = g_{m1}(V_{in} - V_s) + g_{mb1}(0 - V_s) + \frac{V_{out} - V_s}{r_{o1}} + \frac{V_{out}}{r_{o2}}$

$\downarrow \partial v_{in} \quad v_{out} = 0$

since $v_{out} = 0$

$\frac{\partial i_o}{\partial v_{in}} = g_{m1} - g_{m1}\frac{\partial v_s}{\partial v_{in}} - g_{mb1}\frac{\partial v_s}{\partial v_{in}} - \frac{1}{r_{o1}}\frac{\partial v_s}{\partial v_{in}} + 0 = \frac{\partial v_s}{\partial v_{in}}\frac{1}{R_s}$



$\frac{\partial v_s}{\partial v_{in}} = \frac{g_{m1}}{g_{m1} + g_{mb1} + \frac{1}{r_{o1}} + \frac{1}{R_s}}$

$\therefore G_M = \frac{r_{o1}g_{m1}}{r_{o1}R_s(g_{m1} + g_{mb1}) + r_{o1} + R_s}$

$A_v = \frac{v_{out}}{v_{in}} = -G_M R_{out}$

$= \frac{-r_{o1}g_{m1}}{r_{o1}R_s(g_{m1} + g_{mb1}) + r_{o1} + R_s} \times \frac{r_{o1}' r_{o2}}{r_{o1}' + r_{o2}}$

$= -r_{o1}g_{m1} \times \frac{r_{o2}}{r_{o1}' + r_{o2}}$

$= \frac{-r_{o1}r_{o2}g_{m1}}{R_s + r_{o1} + r_{o2} + R_s r_{o1}(g_{m1} + g_{mb1})}$ #